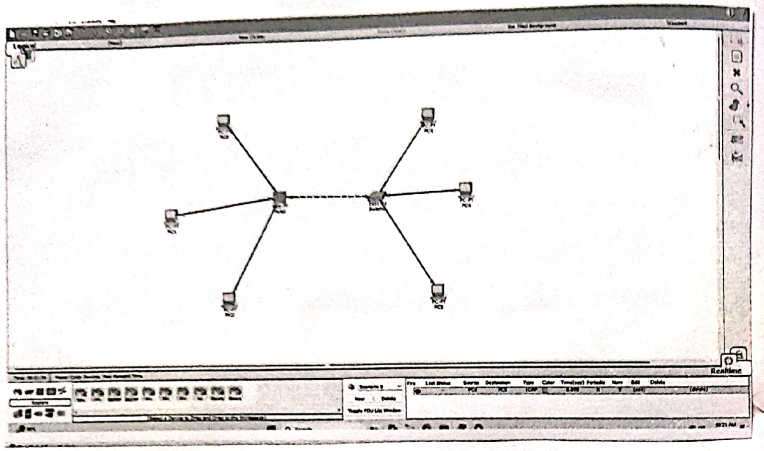
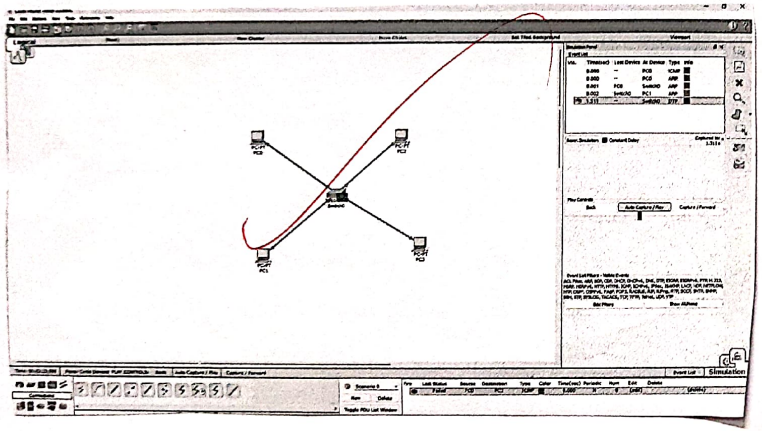




EX 3B · TO ANALYSE BEHAVIOUR OF NETWORK DEVICES USING CISCO PACKET TRACER

AIM · To analyse the behaviour of network devices using Cisco Packet Tracer simulator by designing simple networks and observing data communication.

ALGORITHM:

Case 1 : Using Hub

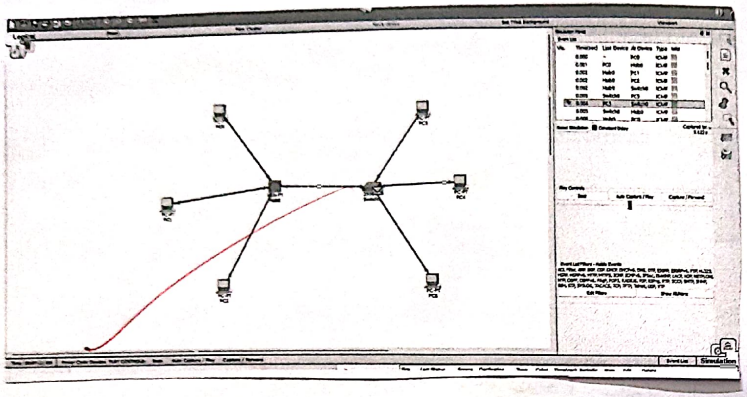
1. Place 1 Hub and connect 4-6 Pc.
2. Configure IP address.
3. Switch to simulation mode
4. Send a simple PDU between 2 Pc's
5. Observe packet transmission.

Case 2 : Using Switch

1. Place 1 switch and connect 4-6 Pc's
2. Configure IP address
3. Send a simple PDU
4. Observe packet Forwarding.

Case 3 : Using Hub and Switch.

1. Place 1 Hub and 1 switch.
2. Connect both devices together
3. Connect Pc's to the Hub and Switch.
4. Configure IP addresses.
5. Send packets between Pc's.
6. Compare the behaviour of Hub and Switch.



PROCEDURE:

1. Open Cisco Packet Tracer.
2. Design:
   - Network using only a Hub
   - Network using only a switch.
   - Network using both Hub and Switch.
3. Assign IP address to all hosts.
4. Enter Simulation Mode
5. Use the Add Simple PDU ~~tool~~ to send packets
6. Observe packet flow and compare functioning of Hub and Switch.

OUTPUT:

Hub:
- broadcasts packets to all connected devices
- More network traffic and collisions

Switch:
- sends packets only to the destination device.
- Faster communication with fewer collisions.

Hub + Switch
- Hub broadcasts packets.
- Switch forwars packets intelligently using MAC address table.

RESULT:

The behaviour of Hub & switch was succesfully analysed using Cisco Packet tracer